Exploratory Data Analysis on Superstore Sales Dataset

1. Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Superstore Sales Dataset using Python. The goal is to identify sales trends, top-performing products, category-wise revenue, and business insights through data visualization.

2. Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook / VS Code

3. Dataset

- Dataset: Superstore Sales Dataset
- Rows: 9800
- Columns: 18

4. Data Inspection

- Loaded the dataset successfully.
- Checked dataset shape, column names, data types, and missing values.
- Observed missing values in the Postal Code column, which was not significant for sales analysis.

5. Exploratory Data Analysis

The following analyses were performed:

- Monthly Sales Trend Analysis
- Top 10 Best Selling Products
- Revenue by Category
- Correlation Heatmap
- Pie Chart for Sales Distribution

6. Key Findings

- Sales varied across different months.
- A few products generated significantly higher sales than others.

- Different product categories contributed differently to total revenue.
- Numerical features showed useful relationships in the correlation heatmap.
- Sales distribution highlighted the contribution of different customer segments/categories.

7. Conclusion

The analysis provided valuable business insights into sales performance and product trends. These insights can help businesses improve inventory planning, marketing strategies, and decision-making. This project also strengthened practical skills in Python, data analysis, and data visualization.